INCH-POUND

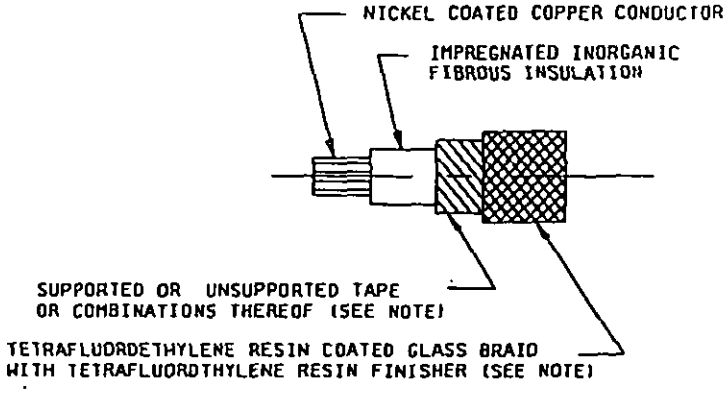
MIL-W-25038/10
16 Sep 1993
SUPERSEDING
MIL-W-25038/1C
20 SEPT 1991

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, HIGH TEMPERATURE,
FIRE RESISTANT, FLIGHT CRITICAL, NORMAL WEIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-25038.



NOTE: THE OUTER TWO LAYERS MAY BE REVERSED IN THEIR SEQUENCE. THE REVERSAL IS OPTIONAL.

1/ The outer two layers may be reversed in their sequence. The reversal is optional.

FIGURE 1. Wire, electrical.

REQUIREMENTS:

Design and construction: See figure 1.

Temperature rating: 260°C.

Voltage rating: 600 V rms.

Color: Unless otherwise specified by the Part or Identifying Number (PIN), the color shall be white.

Changes to the text have been highlighted.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Thermal shock: Maximum change in insulation measurement:

0.060 inch for sizes 22 through 10;
0.125 inch for sizes 8 through 1;
0.187 inch for sizes 0 through 0000.

Durability of identification: 250 cycles (500 strokes) with 150 gram weight.

PIN: M25038/1- (wire size from table I)-(color code designator in accordance with MIL-STD-681).

Examples:
Size 20, white - M25038/1-20-9;
Size 20, white with orange stripe - M25038/1-20-93.

NOTE: Materials used in construction of the wire must be identified and approved by the qualifying activity. Wire construction cannot contain asbestos.

TABLE I. Finished wire construction and performance details.

Wire size 1/	Conductor stranding			Maximum conductor diameter (inch)	Maximum dc resistance 1,000 feet at 20°C (ohms)	Finished wire diameter (inch)		Weight lbs/1,000 feet (max) pounds	Performance details	
	Number of strands	AWG size	Nominal diameter (inch)			Min	Max		Bend test mandrel	
									OD (inches)	Load (pounds)
22	19	34	.0063	.033	23.70	.100	.116	10.0	6	2
20	19	32	.008	.041	14.60	.109	.125	12.0	6	2
18	19	30	.010	.052	9.14	.119	.135	15.0	6	2
16	19	29	.0113	.061	6.85	.127	.147	19.0	8	3
14	19	27	.0142	.074	4.32	.150	.170	25.0	8	3
12	19	25	.0179	.093	2.78	.165	.185	35.0	8	3
10	49	27	.0142	.128	1.68	.210	.230	55.0	10	5
8	133	29	.0113	.176	.936	.256	.280	85.0	12	5
6	133	27	.0142	.218	.591	.318	.342	127	14	10
4	133	25	.0179	.272	.375	.383	.407	192	16	10
2	665	30	.0100	.345	.241	.460	.484	291	20	15
1	817	30	.0100	.384	.196	.497	.533	347	22	15
0	1045	30	.0100	.432	.153	.537	.573	415	24	20
00	1330	30	.0100	.490	.120	.595	.635	520	24	25
000	1672	30	.0100	.548	.096	.660	.700	648	24	30
0000	2109	30	.0100	.615	.077	.730	.770	793	24	30

1/ Wire size 22 through 12 are inactive for new design. For new design use MIL-W-25038/3.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - OS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-2032-1)

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